



DEALER BEST PRACTICE SERIES

Sealed Storage / Staging
Cabinets & Short Block
Kits Reduce Costs

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Sealed Storage / Staging Cabinets Reduce Costs 0

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1.0 Introduction

Great attention is given today to maintaining the cleanliness of parts, both new and reused, throughout the rebuild process. The most common method to protect parts in staging is using plastic wrap around a storage box containing the parts or shrink-wrapping the individual parts themselves. New parts are typically delivered to the workstation in their factory packaging, so the technician must do the unwrapping and waste disposal of the packaging materials.

An alternative method of protecting parts is the growing use of sealed storage cabinets. This reduces costs in several ways:

- Far less consumption of plastic wrap and shrink-wrap materials.
- Time spent in wrapping and unwrapping parts significantly reduced.



Connecting Rod Cabinet – Rods organized by work order number.

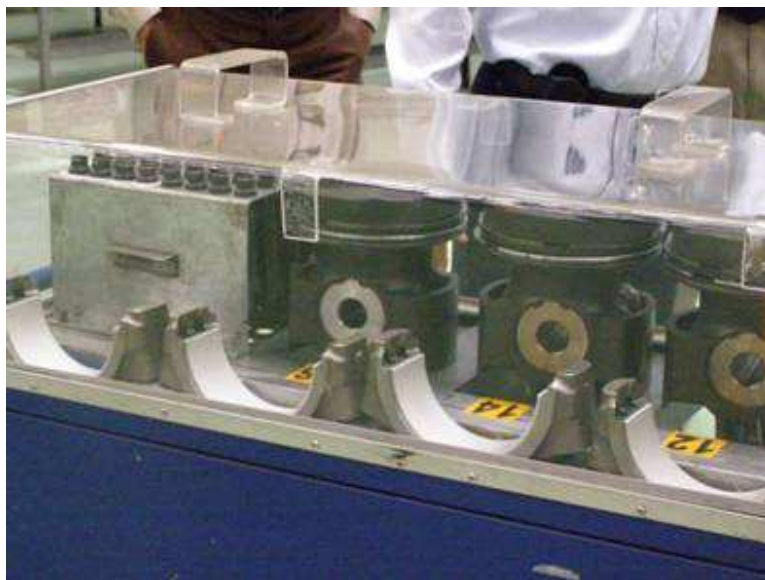
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2.0 Best Practice Description

There are many possibilities for the application and use of sealed storage / staging cabinets.

- To deliver parts to a workstation
- Staging parts in process
- Factory-like workflow

Mobile cabinets



Process

- Cart labeled with work order number.
- Parts are organized by cylinder number.
- Short block technician does not have to unwrap parts.
- All the parts are organized and accessible.
- Cart stored in a staging area & delivered to work bay at the moment it is required

Cart

- Top of cart is Plexiglas two-piece lid with handles.
- Sides of cart are heavy plastic mounted with snaps and Velcro and zippers for access.
- Cart is mobile, on heavy wheels
- Cart has VCI cartridge to protect from rust.
- Note snap rings are all oriented in same position for easy Quality Control inspection
- Locations numbered

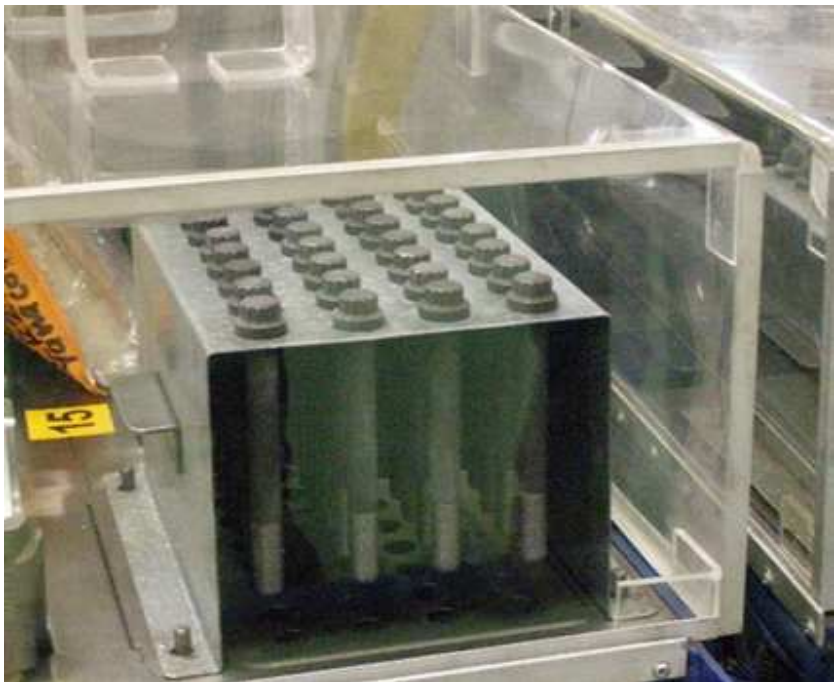
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Photos from Ferreyros

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Ferreyros has both mobile and fixed camshaft storage cabinets. Dealer has started using a reusable soft canvas “bag” to protect the camshaft from contamination instead of the traditional shrink-wrap plastic.



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Fixed Cabinets

Fixed cabinets can be permanently located in a work bay or in a general staging area.



The above photos are from CI Walker. Cabinets have plexiglass doors that open and close for access. These cabinets provide the most efficiency if located between the piston, rod, liner rebuild area and the general short block or general engine assembly bays. This placement eliminates one stage of handling and greatly reduces incidental handling damage exposure. Parts are loaded through the back and retrieved for final assembly through the front.



The above photos are from Ferreyros – fixed storage cabinets.

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Covers

Permanent covers can be used to replace tarps or plastics.



Wheel station bearing preload tooling is protected from contamination with permanent Plexiglas covers.

Corrosion protection



VCI supplies to protect parts from corrosion are often used with the sealed cabinets.

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3.0 Implementation Steps

- Identify areas in the shop where disposable plastic or shrink-wrap is being used to protect parts in process.
- Identify areas in shop where skilled technicians spend time unwrapping or unpacking parts.
- Compare cost of consumables to cost of permanent storage cabinets.
- Design / fabricate cabinets.

4.0 Benefit

- Rebuild cleanliness standards are maintained.
- Parts in process are kept organized.
- Efficient use of skilled assembly technician time
- No lost time wrapping, unwrapping, or searching for parts
- Effective use of out-of-bay staging and floor space
- Less cost of consumables.

5.0 Resources Required

- Metal shop
- Dealer fabricated costs for cabinets can vary from approximately \$800 for a simple design to \$4000 for a large high capacity cabinet.

6.0 Supporting Attachments / References

7.0 Related Best Practices

- 0408-4.1-1116 - Short Block Kits Save Time, Improve Quality

8.0 Acknowledgements

This Best Practice was written by:

Dale Brehm
Caterpillar Global Mining
Brehm_Dale_E@cat.com

Recognition: Ferreyros SAA
Aninu Mitacc Vizcardo
amitacc@ferreyros.com.pe

Photos from Ferreyros and CI Walker.

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